Causal Inference and Uplift Modeling for Online Advertising

Executive Summary

This report presents a comprehensive causal analysis of the Criteo uplift dataset to evaluate the incremental effectiveness of online advertising. Unlike traditional predictive modeling approaches that focus on identifying users likely to convert, this analysis emphasizes **incremental impact**, specifically how advertising changes user behavior relative to a control condition.

Three key hypotheses are investigated:

(1) whether advertising primarily drives engagement or purchasing behavior,

(2) whether treatment effects vary across user segments, and

(3) whether uplift-based targeting strategies outperform a treat-all approach.

The findings reveal that while advertising increases both site visits and conversions, its overall effect is modest and highly uneven across the population. A small subset of users contributes disproportionately to incremental gains. Furthermore, uplift-based targeting strategies significantly outperform both random targeting and treat-all policies, suggesting that data-driven deployment can dramatically improve efficiency.

Hypothesis 1: Advertising Impact Across Funnel Stages

<u>Summary(control → treated)</u>

1) Visit rate

- Control: 3.82%
- Treated: 4.85%
- ATE on visit = **+1.034 percentage points**
- Meaning: showing an ad results in approximately **1 additional user out of every 100** visiting the site

2) Overall conversion rate

- Control: 0.19%
- Treated: 0.31%
- ATE on conversion = **+0.115 percentage points**
- Meaning: advertising generates approximately **0.115 additional conversions per 100 users exposed**
- This corresponds to roughly **1 additional conversion per ~870 users shown an ad**

3) Conversion given visit (conversion | visit)

- Control: 5.07%
- Treated: 6.36%
- ATE on conversion|visit = **+1.292 percentage points**
- Meaning: **conditional on visiting the site**, treated users exhibit a **higher likelihood of purchasing**

<u>The key insight</u>

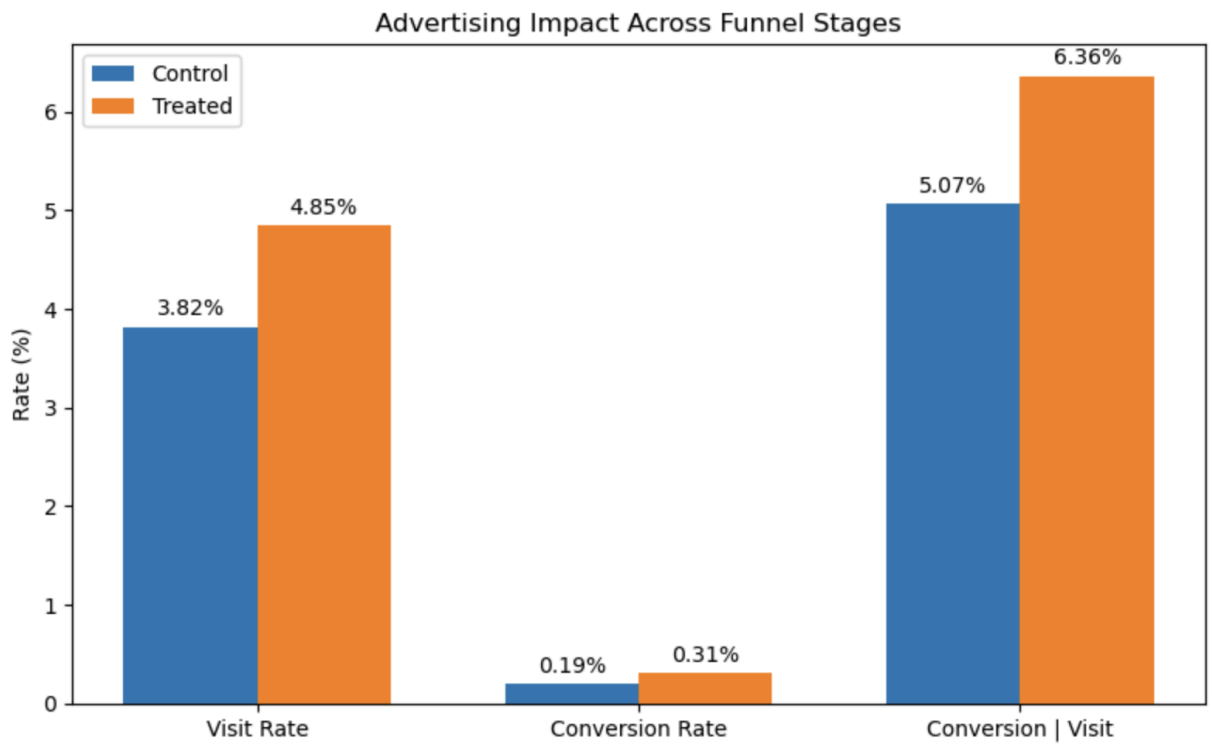
The advertising intervention influences both stages of the user funnel, but not equally:

- The primary effect is at the awareness stage (visits), indicating that **advertising is effective in driving engagement**
- A secondary effect exists at the conversion stage, where **treated users who visit are more likely to purchase**
- However, due to the extremely low baseline conversion rate, the absolute conversion gain remains small

Below are the key outcome metrics and treatment effects, along with a visualization illustrating how advertising impacts each stage of the funnel:

```
Visit rate (C, T): 0.03820095691954503 0.048543360048743316
Conversion rate (C, T): 0.0019375880152813366 0.0030894610674129645
Conversion|Visit (C, T): 0.05072092878097497 0.06364332968115058

ATE Visit: 0.010342403129198284 (1.034 pp)
ATE Conversion: 0.0011518730521316279 (0.115 pp)
ATE Conversion|Visit: 0.012922400900175608 (1.292 pp)
```



Implications

The results show that **advertising primarily influences the visit stage**, with a much larger effect on engagement (+1.034 pp) than on overall conversion (+0.115 pp). This suggests that **ads are more effective at bringing users to the site than directly converting them**.

This **refutes the common assumption that advertising mainly drives purchases**. Instead, it highlights that conversion improvements depend not only on exposure but also on what happens after the user arrives on the site.

Next Steps

- **Improve post-click experience** (landing pages, UI, pricing) to convert increased traffic into purchases
- **Segment users based on visit behavior** to understand where conversion drop-off occurs
- **Combine funnel analysis with uplift modeling** to identify users who are both likely to visit and convert

Broader Impacts

- **Focusing only on driving traffic may lead to inefficient systems** that prioritize clicks over meaningful outcomes.

- From a business perspective, this can **inflate engagement metrics without improving revenue**.
- From a user perspective, **excessive exposure to ads without meaningful value may degrade experience and trust**.

Hypothesis 2: Heterogeneous Treatment Effects

Summary

The uplift-by-decile analysis reveals that **advertising effectiveness varies substantially across user segments**. The **highest predicted uplift group (Decile 1) achieves a realized uplift of approximately 0.575 percentage points**, which is about **five times larger** than the overall average treatment effect (0.115 pp).

In contrast, **most deciles show uplift values close to zero**, indicating that for the majority of users, advertising has little to no effect on conversion. **A small segment even exhibits slightly negative uplift**, suggesting that advertising **may be ineffective or potentially counterproductive for some users**.

Some irregularities, such as the higher uplift observed in **Decile 10, likely reflect noise or limitations in model ranking at the extremes**.

What the T-Learner Does

To estimate these differences, we use a T-learner, which trains two separate models:

- **One on treated users** (those who saw the ad)
- **One on control users** (those who did not)

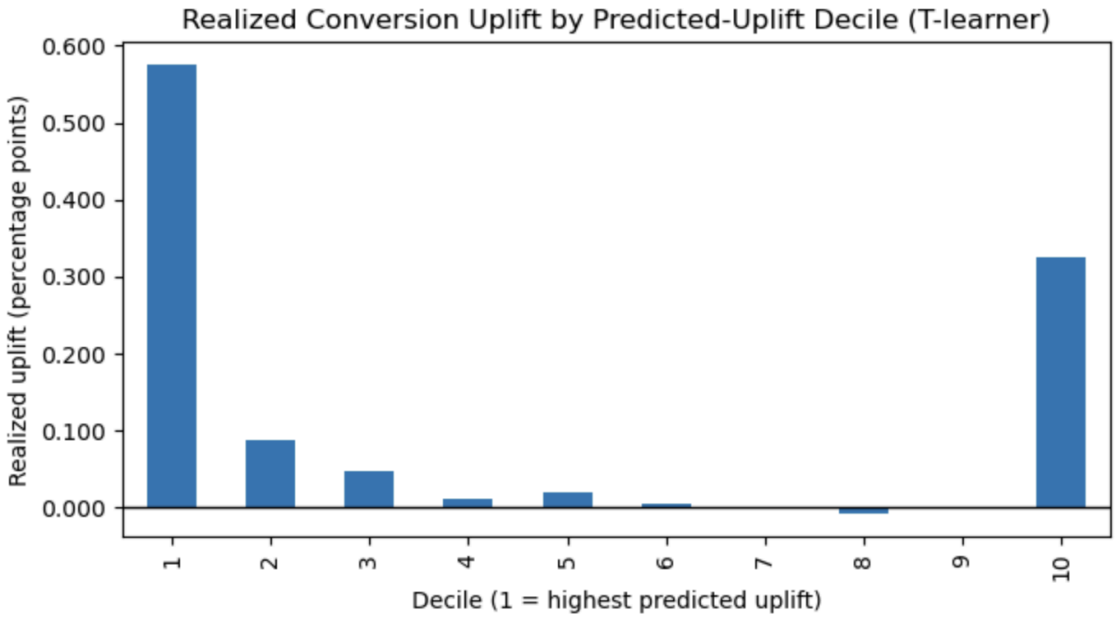
Each model predicts the probability of conversion under its respective condition. The difference between these predictions gives an **estimate of individual uplift**, representing how much the ad changes a user's likelihood of converting.

Users are then ranked by predicted uplift and **grouped into deciles** to evaluate how well the model identifies high-impact segments.

The key insight

The impact of advertising is **not uniform across users**. Instead, it is **highly concentrated in a small subset of users**, commonly referred to as **"persuadables."**

Below are the computed realized uplift values for each decile, together with a plot illustrating how advertising impact differs across segments ranked by predicted uplift:




```
Realized uplift by decile (percentage points):
decile
1     0.575
2     0.087
3     0.048
4     0.011
5     0.019
6     0.006
7     0.000
8    -0.008
9     0.002
10    0.325
dtype: float64
```

Implications

These findings **support the hypothesis that treatment effects are heterogeneous** and highlight **the inefficiency of treating all users equally**. A large portion of advertising exposure is effectively wasted on users who do not benefit, while a smaller segment drives most of the incremental value.

Next Steps

- Improve model stability and ranking consistency
- Explore alternative uplift methods (e.g., X-learner, DR-learner)
- Analyze feature importance to better understand drivers of high uplift

Broader Impacts

Segment-based targeting can significantly improve efficiency but must be implemented carefully. **Over-targeting specific groups may introduce bias or unequal exposure**, so **fairness and transparency should be considered** when deploying uplift-based strategies.

Hypothesis 3: Policy Evaluation and Targeting Efficiency

Summary

The policy evaluation results show that **uplift-based targeting significantly outperforms both random targeting and a treat-all strategy**. The **Qini curve demonstrates strong gains in the early targeting range**, indicating that the model **effectively ranks users by their responsiveness to advertising**.

For example, **targeting the top 10% of users yields approximately 1390 incremental conversions**, compared to only **about 277 conversions under random targeting**. Targeting the **top 20% increases this to around 1710 conversions**, while treating all users yields approximately 2771 conversions.

Notably, **targeting only the top 10% of users captures roughly 50% of the total incremental conversions**, despite using only a small fraction of impressions.

What the Qini Table shows

This table shows how many **incremental conversions** are achieved when targeting different percentages of users ranked by predicted uplift. Each row compares **uplift-based targeting** with a **random baseline**, allowing us to see how much better the model performs.

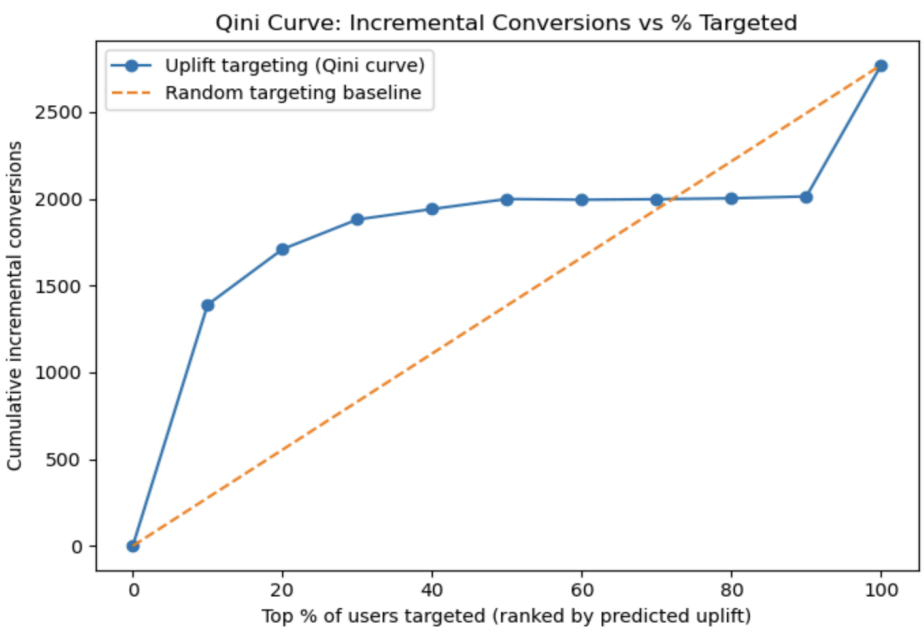
- Targeting the top 10–20% of users produces **significantly higher incremental conversions** than random targeting, indicating strong model effectiveness.
- The **Qini (over random)** values are highest at lower targeting percentages, showing that the model is best at identifying high-value users early.
- As the targeting percentage increases (e.g., 70–90%), the advantage over random decreases and can become negative, meaning lower-ranked users add little or no value.
- Overall, the results demonstrate that **advertising impact is concentrated in a small subset of users**, supporting targeted strategies rather than treating all users.

	Top % targeted	Cumulative incremental conversions	Random baseline	Qini (over random)
0	10	1389.959644	277.110018	1112.849626
1	20	1709.689846	554.220035	1155.469811
2	30	1880.994701	831.330053	1049.664648
3	40	1941.283431	1108.440070	832.843361
4	50	1999.245247	1385.550088	613.695159
5	60	1994.836624	1662.660105	332.176519
6	70	1998.086205	1939.770123	58.316082
7	80	2003.239873	2216.880140	-213.640267
8	90	2013.991721	2493.990158	-479.998437
9	100	2771.100175	2771.100175	0.000000

What the Qini Curve Shows

The **Qini curve represents the cumulative incremental conversions achieved when targeting users ranked by predicted uplift**, from highest to lowest.

- The **blue curve** represents **uplift-based targeting**
- The **orange dashed line** represents **random targeting**
- The **large gap between the blue curve and the random baseline at low percentages** (e.g., top 10–20%) indicates that the model is **effectively identifying high-value users**.
- However, **as targeting expands beyond the top segments**, the curve **begins to flatten, reflecting diminishing returns**.



Concept of Policy (from the Qini Curve)

The policy is defined by **ranking users based on predicted uplift** and selecting the top percentage of users to target. The Qini curve visualizes how different policies perform by showing the **cumulative incremental conversions achieved** as we target more users:

- Moving along the x-axis (e.g., 10%, 20%, 30%) represents **different targeting policies**
- Each point corresponds to a policy like:
    - "Target top 10% of users"
    - "Target top 20% of users"
    - "Target all users"
- A **good policy** is one that achieves a high number of incremental conversions **with fewer users targeted**, meaning it is more efficient.

	Policy	Estimated incremental conversions
0	Target top 5%	1098.016675
1	Target top 10%	1389.970807
2	Target top 20%	1709.653564
3	Target top 30%	1880.998900
4	Target top 50%	1999.247805
5	Target top 100%	2771.100175
6	Treat-none (0%)	0.000000

Key Insight

- Targeting the **top 10% achieves ~50% of the total incremental conversions**
- This demonstrates that most of the advertising impact is **concentrated in a small subset of users.**

Implications

The results strongly support the hypothesis that **uplift-based targeting is more efficient than a treat-all strategy**. The Qini curve shows that most of the incremental value is captured when targeting the highest-ranked users, and the policy table confirms that a small portion of users contributes disproportionately to total conversions.

For example, targeting the top 10% of users achieves approximately **50% of the total incremental conversions**, demonstrating that broad exposure is not necessary to achieve strong performance. This challenges the common assumption that increasing coverage always improves outcomes. Instead, it highlights that **precision in targeting is more important than scale**.

Overall, these findings indicate that **organizations can significantly reduce advertising costs** while maintaining effectiveness by focusing on high-uplift users.

Potential actions from this findings for businesses:

- Select an **optimal targeting threshold** (e.g., top 10–20%) based on business objectives and budget constraints
- Incorporate **cost and revenue metrics** (cost per impression, value per conversion) to evaluate profit, not just conversions
- Validate the targeting policy through **online A/B testing** before deployment

Broader Impacts

Adopting uplift-based targeting can improve efficiency by reducing unnecessary ad exposure, which may lead to better user experience and lower operational costs. However, **selective targeting also raises important ethical considerations**.

Focusing only on high-response users may unintentionally **exclude certain groups**, potentially reinforcing existing behavioral or demographic biases. Therefore, it is important to incorporate **fairness checks and transparency** when deploying such policies to ensure that targeting decisions do not lead to unintended inequality or exclusion.